

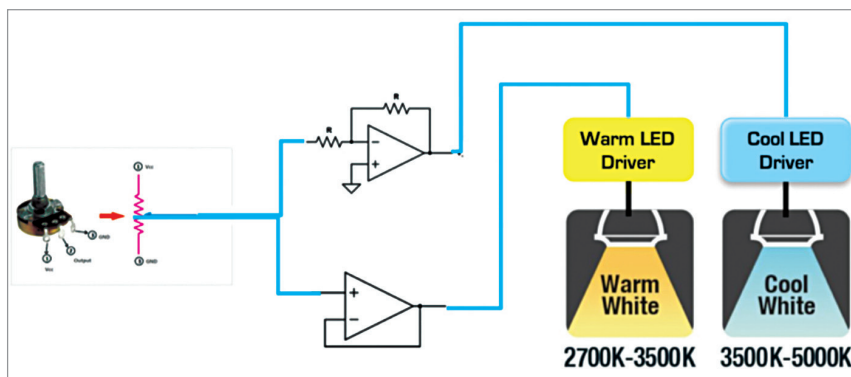


Fig. 4: Warm and cool LEDs used for producing human-centric light

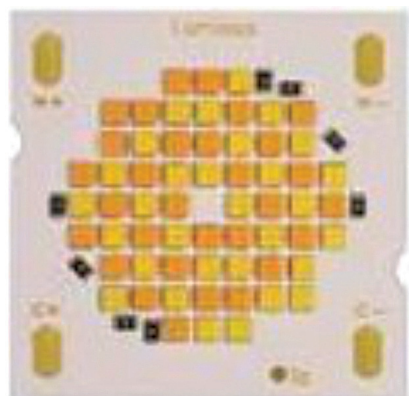


Fig. 5: Module with alternate cool and warm LEDs

level), will have a positive impact on our bio rhythm or circadian rhythm.

Unlike the conventional bulbs, an LED light source can be controlled by the current flowing through it. Higher the current, higher the intensity! This ability to control the intensity of light helps the growth of human-centric lighting business.

Timing

Third aspect that impacts us, along with colour and intensity, is the time when the lights are switched on and off. Adjusting the intensity of the light, along with the colour at different times of a day, can create a healthy environment for us due to impact of light on our circadian rhythm. For example, excess light in the evening can reduce the secretion of the melatonin and thus lead to lack of sleep, which can cause multiple health issues.

Using LEDs for a healthy life

With the background on the circadian rhythm and characteristics of natural light let us learn how lights can be used to impact human biological

rhythm in a positive way. Human-centric lights became popular with the emergence of LED lights and their ability to control the colour, intensity, and timing with ease.

The top-most factor that impacts the human-centric lighting is the spectrum of the light as colour is critical for impacting the circadian rhythm. Change of colour was difficult with conventional light sources like incandescent bulbs and compact fluorescent lamps. This challenge was overcome once the LED based light sources became common, accelerated by the fact that prices of LED lights have bottomed out. This has helped in building low-cost human-centric lighting systems that impact our circadian rhythm positively, leading to a healthier life.

In human-centric lighting, first blue-rich, high-intensity light is produced in the early morning until midday, tapering off towards 4pm in the evening. After 4pm, only 3,000 degrees Kelvin light (typically warm white) is produced, which is reduced in intensity until bedtime.

To choose the right LEDs we need to take care of a parameter known as correlated colour temperature (CCT), which defines the colour appearance of a white LED in degrees Kelvin (K). A warm light is around 2700K, moving to neutral white at around 4000K, while a cool white is at 5000K and beyond.

Basics of human-centric lighting

Let us now see the basic architecture of human-centric lighting system, which is based on a mixture of cool white and warm white lights. It is achieved using a simple PWM control with cool white and warm white

LEDs that are driven by inverted PWM waveforms. This essentially means when the cool white LED is glowing bright and warm, white LED will be glowing dim (in proportion to the PWM on and off times).

Fig. 4 shows how both types of LEDs (warm and cool white) are connected to generate human-centric lighting with a single control. Readers can experiment with a simple 555 timer based circuit to see how the system works. Please note, the LEDs are always driven by current sources and not by voltage sources. This means the PWM circuit should drive the current sources.

However, one challenge with PWM-driven LED lighting systems is that when a video is taken horizontal lines appear in the video. This is due to the LEDs that are switched on and off at PWM frequency, which does not synchronise with the camera's frame rate. So to check whether the lights are PWM-driven or linear-current-driven, hold your mobile phone camera in video mode towards the LED light. If you see scrolling lines, the LED light is operating in PWM mode, and it is advisable to use a linear-current-driven LED for video recording.

Second important aspect is how the LEDs are positioned. Cool and warm LEDs are alternated so that the net output is mixed light. We get readymade modules from vendors that can be used directly as these have prepositioned LEDs. One such module is shown in Fig. 5. You can see the way warm and cool LEDs are positioned in the module (yellow and orange ones).

Last important aspect is the use of a diffuser in the light to merge/mix the warm and cool LED lights into one single light source. Today readymade human-centric lighting luminaires are available in different shapes and sizes, either with manual control or automatic control.

Essentials of human-control lighting

Human-centric/circadian rhythm lighting system for a large space like office or home essentially has two components:

1. Human-centric/circadian rhythm lighting modules with controllable drivers